LESSON PLAN

WEAPON GUIDANCE (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	17
Total No. of Lesson Plans	22	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Definition of grounding, various types of grounding, grounding methods
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Definition of grounding, various types of grounding, grounding methods.
5. Specific Objectives	19.1.1.1Trainee will be able to explain aircraft wiring and Causes of wiring degradation.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstrate proper grounding clamp attachment and verify continuity with multimeter; Safety: Ensure power off and PPE worn; Supervision: Instructor monitors tool usage and safety compliance.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of grounding in an aircraft electrical system?
(b) List two common causes of aircraft wiring degradation.
(c) Why is a single-point ground preferred over multiple grounds in certain systems?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of electrical grounding • Path of least resistance to earth • Safety function of grounding systems • Reference to IB Chapter 19	L & Ds	PPT, IB	15 min
(b)	• Single-point grounding method • Multi-point grounding method • Distributed grounding method • Comparison of grounding types	L & Ds	PPT, PDF	20 min
(c)	• Grounding clamp installation procedure • Verification of ground continuity • Common grounding faults identification • Precautions during grounding work	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between single-point and multi-point grounding in aircraft systems.
(b) Describe the steps to verify proper grounding continuity on a live aircraft panel.
(c) Identify three causes of wiring degradation related to improper grounding.

12. Home Assignment and Follow Up	Prepare a brief report comparing grounding and bonding methods as covered in LP 17 and LP 18, highlighting key differences in application and safety implications.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Definition of bonding, various types of bonding, bonding methods.	Refer to the master syllabus reference list	LP 18

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Wing Commander, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min